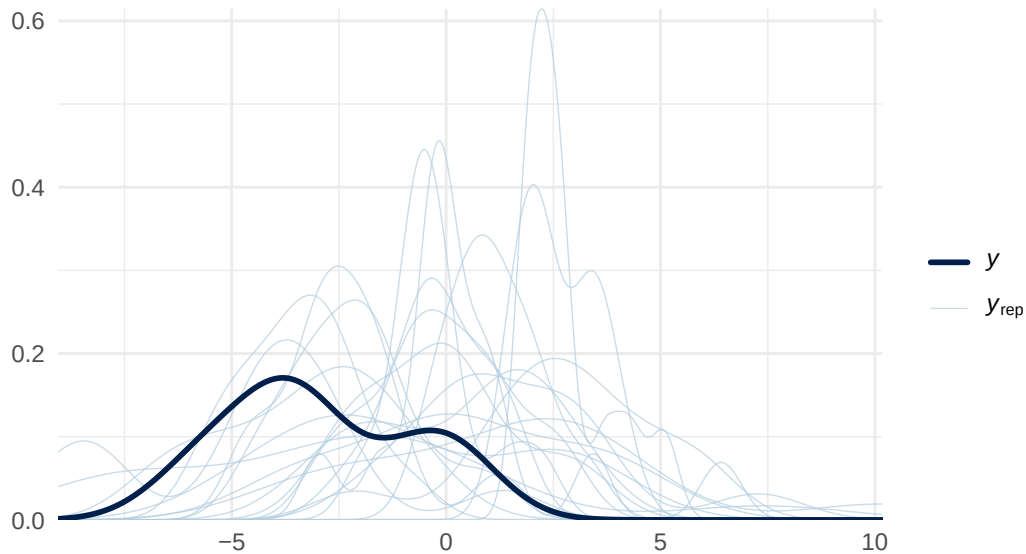


Supplementary File 5

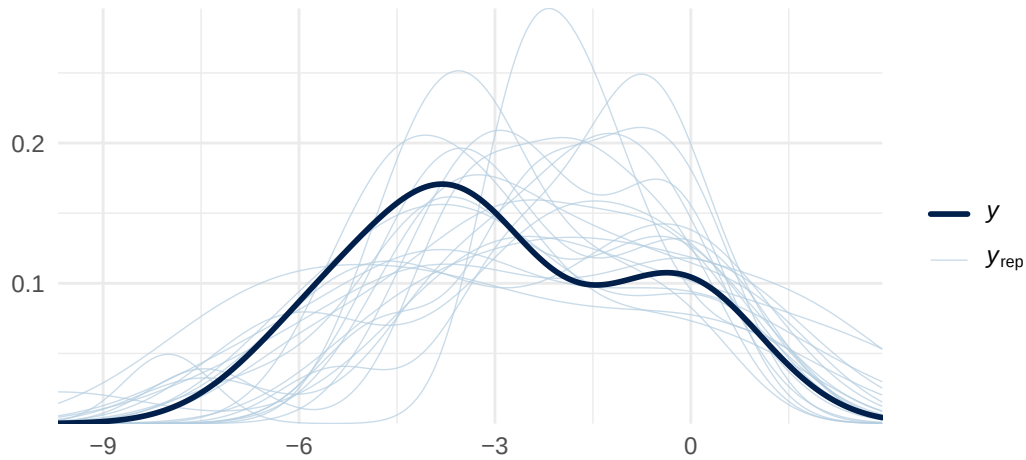
This supplementary file shows diagnostic plots for the Bayesian model and a funnel plot.

5a) Prior predictive check



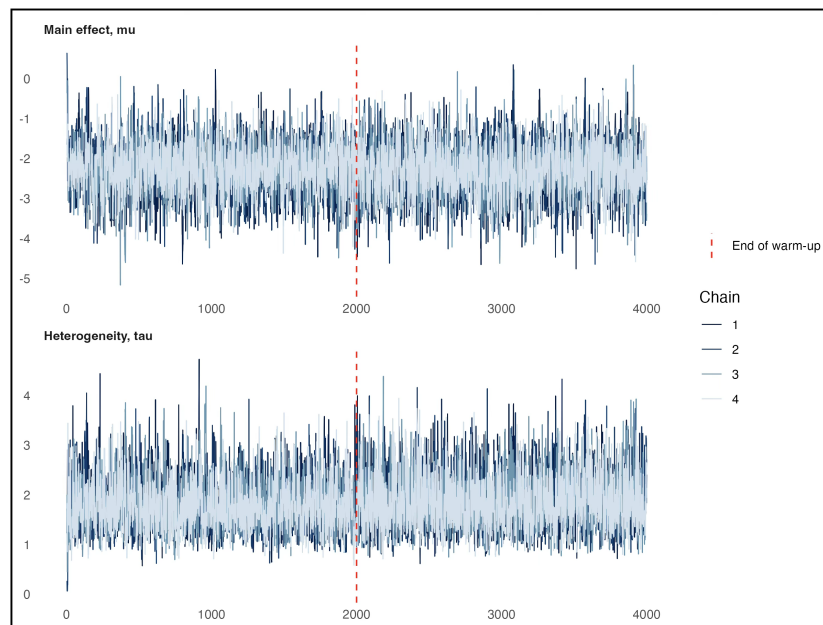
Prior predictive check, using `pp_check()` from the `bayesplot` package. Density of the observed effect sizes (dark line) compared with 20 data sets simulated from the prior predictive distribution (light lines), obtained by fitting the model with `sample_prior = "only"`. This checks whether the priors imply plausible effect sizes before the observed data are taken into account.

5b) Posterior predictive check



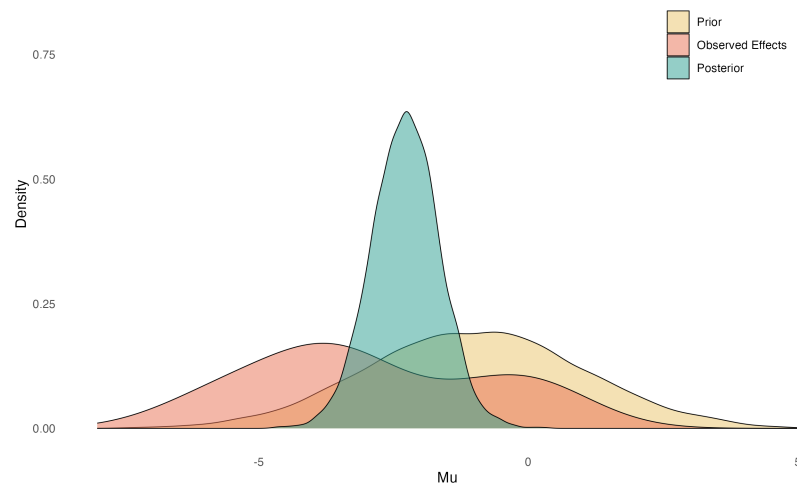
Posterior predictive check, using `pp_check()` from the `bayesplot` package. Density of the observed effect sizes (dark line) compared with 20 data sets simulated from the posterior predictive distribution of the fitted model (light lines). This checks how well the fitted model captures and reproduces the observed data.

5c) MCMC trace plots for mu and tau

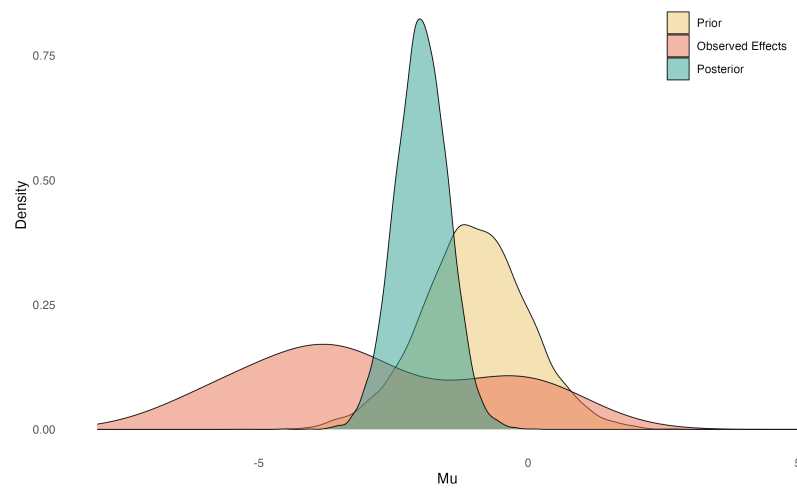


5d) Prior, observed effects and posterior across prior specifications

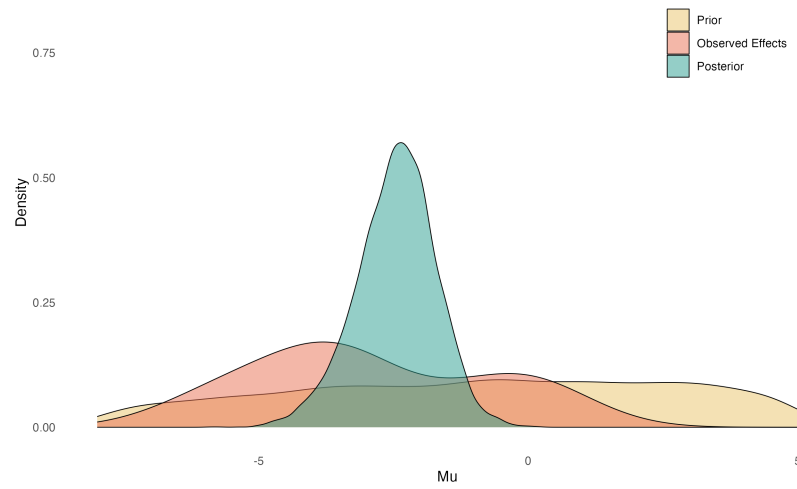
Sensi. analysis: main Prior, Observed Effects, and Posterior of Main effect, μ



Sensi. analysis: narrow Prior, Observed Effects, and Posterior of Main effect, μ



Sensi. analysis: wide Prior, Observed Effects, and Posterior of Main effect, μ



5e) Funnel plot

